

Functions

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Based on the lecture note uploaded on SPeCTRUM

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Understand what is a function and determine if a relation is a function.
- ☐ Determine the domain and range of a function.
- ☐ Determine if a function is one-to-one.
- ☐ Determine if a function is onto.
- ☐ Determine if a function is one-to-one correspondence.
- ☐ Understand what is an invertible function and determine the inverse of a function.
- ☐ Understand what is a composite function.

Functions

Let X and Y be two non-empty sets. A **function** f from X to Y is a rule that assigns a unique element of Y to each element of X .

Note: A function from X to Y is a relation but not all relations from X to Y are functions.

The set X is called the **domain** of f and the set Y is the **codomain** of f , and we write $f : X \rightarrow Y$.

For each $x \in X$, we write $f(x) = y$ if y is assigned to x .

$f(x)$ (read as “ f of x ”) is called the **image of x under f** (or the **value of f at x**).

The set of all such image values is called the **image of f** , and is denoted by $\text{Im}(f)$.

* A relation is a function if each element in the domain maps to one and only one element in codomain.

Functions

Example: Let $A = \{a, b, c, d\}$ and $B = \{1, 2, 3, 4, 5\}$. Which of the following relations are functions map from A to B ?

- ① $R_1 = \{(a, 2), (b, 1), (b, 3), (c, 1), (c, 4), (d, 1)\}$
Not a function, because $b \in A$ is mapped to two elements in B . (1 and 3)
- ② $R_2 = \{(a, 2), (b, 1), (c, 3), (d, 5)\}$ Is a function.
- ③ $R_3 = \{(a, 2), (b, 1), (d, 5)\}$ Not a function. Because $c \in A$ but c does not map to any element of B .

Let $f : \mathbb{N} \rightarrow \mathbb{Z}$ be the function defined by $f(x) = \begin{cases} 2x, & \text{if } x \text{ is odd;} \\ x - 1, & \text{if } x \text{ is even.} \end{cases}$ 2, 6, 10, 14, ... odd starts from 1

Find

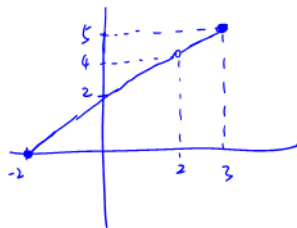
- ① the domain and the codomain of f Domain: $\mathbb{N}$ and Codomain: $\mathbb{Z}$.
- ② the image of 4, the image of 11 and the image of f
 $f(4) = 3$, $f(11) = 22$. $\text{Im}(f) = \{1, 2, 3, 5, 6, 7, 9, 10, 11, 13, 14, 15, 17, 18, 19, \dots\}$

Graph of functions

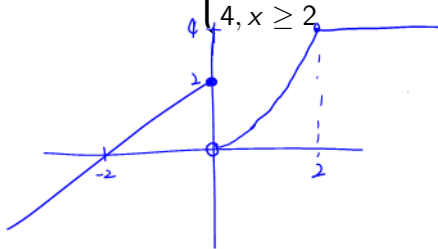
Let f be a function. The **graph** of f consists of all the points (x, y) where $y = f(x)$.

Sketch the graph of the following functions:

$$f(x) = \frac{x^2-4}{x-2} \text{ where } -2 \leq x \leq 3, x \neq 2.$$



$$g(x) = \begin{cases} x+2, & x \leq 0 \\ x^2, & 0 < x \leq 2 \\ 4, & x \geq 2 \end{cases}$$



One-to-one functions

A function $f : X \rightarrow Y$ is said to be **one-to-one** (or **injective**) if and only if for every $y \in Y$, there is at most one $x \in X$ such that $f(x) = y$.

Or equivalently,

$$\forall x_1, x_2 \in X, f(x_1) = f(x_2) \Rightarrow x_1 = x_2.$$

Or equivalently,

$$\forall x_1, x_2 \in X, x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2).$$

To prove a function $f: X \rightarrow Y$ is one-to-one

Let $x_1, x_2 \in X$ such that $f(x_1) = f(x_2)$

$\vdots$

$x_1 = x_2$.

Onto function

A function $f : X \rightarrow Y$ is said to be **onto** (or **surjective**) if and only if

$$\forall y \in Y, \exists x \in X \text{ such that } f(x) = y.$$

In other words, $\text{Im}(f) = Y$ (The image of f is equal to the codomain of f), and we say f is a function **from X onto Y** , or **f maps X onto Y** .

To prove a function $f: X \rightarrow Y$ is onto.

(m1) Let $y \in Y$. Notice that we can always have $x = \underline{\text{(in terms of } y\text{)}} \in X$ such that

$$f(x) = f(\underline{\quad}) = \dots = y.$$

So f is onto.

(m2) Show $\text{Im}(f) = Y$

One-to-one correspondence

A function $f : X \rightarrow Y$ is said to be **one-to-one correspondence** (or **bijjective**) if and only if it is both one-to-one and onto.

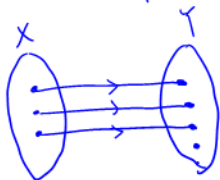
Or equivalently,

$$\forall y \in Y, \exists! x \in X \text{ such that } f(x) = y.$$

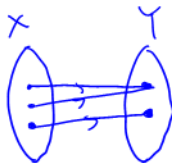
Example: Is each of the following function (i) one-to-one (ii) onto (iii) one-to-one correspondence?

- ① $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x + 1$.
- ② $g : \mathbb{N} \rightarrow \mathbb{R}$ defined by $g(x) = \sqrt{x}$.
- ③ $h : \mathbb{R} \rightarrow \mathbb{R}$ defined by $h(x) = \sin x$.

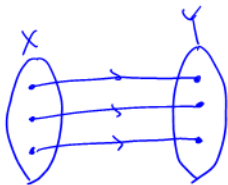
$$f: X \rightarrow Y$$



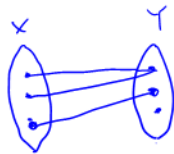
1-1 but not onto



onto but not 1-1



1-1 and onto



neither 1-1 nor onto.

This is to help you understand 1-1 and onto,
you should never use this to justify if a function is 1-1 or onto
formally.

One-to-one correspondence

$f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x + 1$.
domain $\swarrow$ $\nwarrow$ *codomain*

One-to-one

Let $x_1, x_2 \in \mathbb{R}$ such that $f(x_1) = f(x_2)$. Then, $x_1 + 1 = x_2 + 1$ and hence, $x_1 = x_2$. Therefore, f is one-to-one.
domain $\swarrow$

Onto

Let $y \in \mathbb{R}$. We can always have an $x = y - 1 \in \mathbb{R}$ such that $f(x) = f(y - 1) = (y - 1) + 1 = y$. Therefore, f is onto.
codomain $\swarrow$ *always defined in the domain*

One-to-one correspondence

Since f is one-to-one and onto, f is one-to-one correspondence.

One-to-one correspondence

$\{1, 2, 3, 4, \dots\}$ $\leftarrow$ all real number.

$g: \mathbb{N} \rightarrow \mathbb{R}$ defined by $g(x) = \sqrt{x}$.

One-to-one

Let $x_1, x_2 \in \mathbb{N}$ such that $g(x_1) = g(x_2)$. Therefore, $\sqrt{x_1} = \sqrt{x_2}$ and hence $x_1 = x_2$. So, g is one-to-one.

Onto

there is no $x \in \mathbb{N}$ such that $g(x) = 0$.
Notice that $0 \in \mathbb{R}$. but there is no element in $\mathbb{N}$ that maps to 0 under the function g . So, g is not onto.

One-to-one correspondence

Since g is not onto, g is not one-to-one correspondence.

One-to-one correspondence

$h : \mathbb{R} \rightarrow \mathbb{R}$ defined by $h(x) = \sin x$.

One-to-one

Notice that $h(0) = h(\pi) = 0$ but $0 \neq \pi$.

So h is not one-to-one.

Onto

Notice that $2 \in \mathbb{R}$, but there is no $x \in \mathbb{R}$ such that $h(x) = 2$. So, h is not onto.

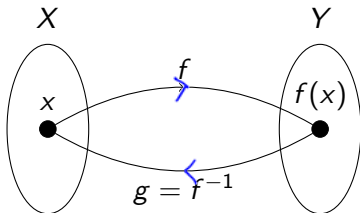
One-to-one correspondence

Since h is neither one-to-one nor onto,
 h is not one-to-one correspondence.

Invertible functions

Let $f : X \rightarrow Y$ be a function. We say f is **invertible** if there exists a function $g : Y \rightarrow X$ such that $\forall x \in X$ and $\forall y \in Y$, $f(x) = y \Leftrightarrow g(y) = x$. We call such a function g an **inverse** of f , and we denote g by f^{-1} .

Note: f^{-1} here does not mean $\frac{1}{f}$.



A function $f : X \rightarrow Y$ is invertible if and only if f is one-to-one correspondence.

Invertible functions

Example: Which of the following functions are invertible?

- ① $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2$.

Since $f(1) = f(-1)$, but $1 \neq -1$, so f is not 1-1.

So f is not 1-1 correspondence and f is not invertible.

- ② $g: \{x \mid x \geq 0\} \rightarrow \{x \mid x \geq 0\}$ defined by $g(x) = x^2$.

one-to-one and onto.

So g is one-to-one correspondence and hence g is invertible.

$$g^{-1}: \{x \mid x \geq 0\} \rightarrow \{x \mid x \geq 0\}$$
$$g^{-1}(x) = \sqrt{x}$$

- ③ $h: \mathbb{N} \rightarrow \mathbb{N}$ defined by $h(x) = 2x$.

Not onto.

So h is not 1-1 correspondence and hence h is not invertible.

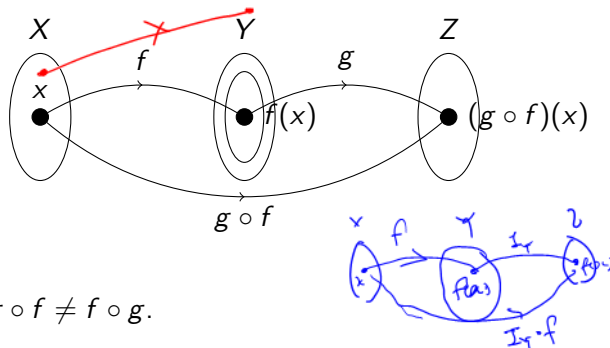
- ④ $k: \mathbb{R} \rightarrow \mathbb{R}$ defined by $k(x) = \sqrt[3]{x}$.

One-to-one and onto

So k is 1-1 correspondence and hence k is invertible.

Composite functions

Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be functions with the property that $\text{Im}(f) \subseteq \text{Dom}(g)$. We define a new function $g \circ f : X \rightarrow Z$ by $(g \circ f)(x) = g(f(x)) \forall x \in X$. This new function $g \circ f$ is called the **composition** of f and g .



In general, $g \circ f \neq f \circ g$.

If $f : X \rightarrow Y$ is any function, then $f \circ 1_X = f = 1_Y \circ f$ where 1_X is the **identity function** on X ($1_X(x) = x, \forall x \in X$).

Composite function

Example: Let $f : \{a, b, c\} \rightarrow \{1, 2, 3\}$ be the function $\{(a, 2), (b, 1), (c, 2)\}$ and let $g : \{1, 2, 3\} \rightarrow \{a, c, d\}$ be the function $\{(1, c), (2, d), (3, a)\}$. Find

- ① $g \circ f$, if exists

$$\text{Im}(f) = \{1, 2\}$$

$$\text{Dom}(g) = \{1, 2, 3\}$$

$$\text{Im}(f) \subseteq \text{Dom}(g)$$

- ② $f \circ g$, if exists

$$\text{Im}(g) = \{a, c, d\}$$

$$\text{Dom}(f) = \{a, b, c\}$$

$$\text{Im}(g) \not\subseteq \text{Dom}(f)$$

So $f \circ g$ does not exist.

So $g \circ f$ exists.

$$g \circ f : \{a, b, c\} \rightarrow \{a, c, d\}$$

$$g \circ f = \{(a, d), (b, c), (c, d)\}$$



$$\begin{aligned} g \circ f(a) &= g(f(a)) \\ &= g(2) = d \end{aligned}$$

Composite function

Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be two functions. Prove that if f and g are one-to-one correspondence, then $g \circ f$ is one-to-one correspondence.



$g \circ f : X \rightarrow Z$ f is one-to-one and onto $f : X \rightarrow Y$
 g is one-to-one and onto. $g : Y \rightarrow Z$

(1-1)
Let $x_1, x_2 \in X$ such that $g \circ f(x_1) = g \circ f(x_2)$
 $g[f(x_1)] = g[f(x_2)]$
 $f(x_1) = f(x_2)$ since g is 1-1
 $x_1 = x_2$ since f is 1-1.
So $g \circ f$ is 1-1.

(Onto)
Let $z \in Z$. Since g is onto, we can always find an $y \in Y$ such that $g(y) = z$. Since f is onto, we can always find an $x \in X$ such that $f(x) = y$. We can take this x and $g \circ f(x) = g(f(x)) = g(y) = z$ and hence $g \circ f$ is onto.

Therefore, $g \circ f$ is a one-to-one correspondence